

3W/6W AUDIO AMPLIFIER Using TDA2003

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This is a simple audio-frequency (AF) amplifier using the popular audio amplifier IC TDA2003. The IC comes in a 5-pin TO-220B package.

Circuit and working

Fig. 1 shows the circuit diagram of the 3W/6W AF amplifier built around IC TDA2003 (IC1), an 8-ohm, 6-watt speaker (LS1) and a few other components.

The amplifier IC delivers 3W output power using a 6V, 500mA power supply, and 6W output power using a 12V, 500mA power supply, with an 8-ohm, 6-watt speaker.

Pin 1 of TDA2003 is the input terminal, which is connected to ground through a 100-kilo-ohm resistor. The audio frequency (or audio signal) is fed to pin 1 of IC1 through the combination of a 10-kilo-ohm potmeter (VR1) and capacitors C1 and C2. Potmeter VR1 is used as volume control. Pin 3 of IC1 is connected to ground. Pin 4 is the output terminal,

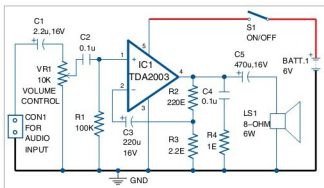


Fig. 1: Circuit diagram of 3W to 6W audio amplifier using TDA2003

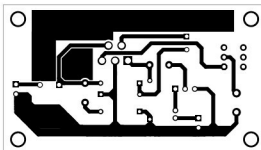


Fig. 2: Actual-size PCB layout of 3W to 6W audio amplifier using TDA2003

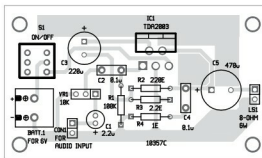


Fig. 3: Components layout for the PCB

which is connected to one terminal of the speaker through a 470µF, 16V capacitor (C5). It is also connected to ground through a 0.1µF capacitor (C4) and a 1-ohm resistor (R4).

Pin 5 of IC1 is connected to +6V power supply via switch S1. Power supply can be provided through a

6V battery or a 6V DC adaptor. Use of a suitable heatsink is recommended for IC1.

Construction and testing

An actual-size, single-side PCB for 3W/6W audio amplifier using TDA2003 is shown in Fig. 2

and its components layout in Fig. 3. After assembling the circuit on the PCB, enclose it in a suitable box.

Solder TDA2003 and other components using a 25W soldering iron. Use a 2-pin connector for input and output connectors, to make your prototype safe and clean.

For testing the circuit, connect a 6V battery to the circuit. Also, connect an 8-ohm, 6-watt speaker to LS1.

If a 12V DC supply is used, voltage ratings of C1, C3 and C5 should be 25V or above.

Take a metal screwdriver and gently touch at input pin 1 of IC1. If your circuit is wired properly, you will hear a humming sound from the speaker. Else, vary potmeter

VR1 to increase the volume until humming sound is heard from the speaker. Now your circuit is ready to use. **EFY**



Raj K. Gorkhali is a regular contributor to EFY and has a keen interest in designing electronic circuits

PARTS LIST

Semiconductors:

IC1 - TDA2003 audio amplifier

Resistors (all 1/4-watt, ±5% carbon):

- R1 - 100-kilo-ohm
- R2 - 220-ohm
- R3 - 2.2-ohm
- R4 - 1-ohm
- VR1 - 10-kilo-ohm potmeter

Capacitors:

- C1 - 2.2µF, 16V electrolytic
- C2, C4 - 0.1µF ceramic disk
- C3 - 220µF, 16V electrolytic
- C5 - 470µF, 16V electrolytic

Miscellaneous:

- CON1 - 2-pin connector
- LS1 - 8-ohm, 6-watt loudspeaker
- S1 - On/off switch
- BATT.1 - 6V battery
- 2-pin terminal connector for battery
- Heat-sink for TDA2003
- Audio signal source
- 6V/12V, 500mA DC power supply (optional)